ITDS 124 Mathematics of Computing

Chapter 3. Counting

3.3 Pigeonhole Principle

หลักรังนกพิราบ

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Pigeons in holes

- Here there are $n = 10$ pigeons in $m = 9$ holes. Since 10 is greater than 9, the pigeonhole principle says that at least one hole has more than one pigeon. (The top left hole has 2 pigeons.)

https://en.wikipedia.org/wiki/Pigeonhole_principle

Pigeonhole Principle

- Another proof technique, one that often makes use of the counting methods.
- **Theorem 1** : **If n pigeons are assigned to m pigeonholes, and $m < n$, then at least one pigeonhole contains two or more pigeons.**

Proof Use proof by contradiction.

Suppose each pigeonhole contains at most 1 pigeon. Then at most m pigeons have been assigned. But since $m < n$, not all pigeons have been assigned pigeonholes. This is a contradiction. At least one pigeonhole contains two or more pigeons.

Existence proof vs Constructive proof

- **Existence proof** (การพิสูจน์ว่ามี)
  - The pigeonhole principle provides an existence proof.
  - There must be an object or objects with a certain characteristic. In Example 1, this characteristic is having been born on the same day of the week.
  - The pigeonhole principle guarantees that there are at least two people with this characteristic but gives no information on identifying these people. Only their existence is guaranteed.
- **Constructive proof** (การพิสูจน์ในเชิงก่อ)
  - A constructive proof guarantees the existence of an object or objects with a certain characteristic by actually constructing such an object or objects.
  - For example, we could prove that given two rational numbers p and q there is a rational number between them by showing that $\frac{p+q}{2}$ is between p and q.

Pigeons and Pigeonholes

- In order to use the pigeonhole principle, we must identify
  - pigeons (objects)
  - pigeonholes (categories of the desired characteristic)

  And be able to count the number of pigeons and the number of pigeonholes.

Example 1

- If eight people are chosen in any way from some group, at least two of them will have been born on the same day of the week.

Solution

- Pigeons :
- Pigeonholes :

- By pigeonhole principle (PHP) :

Example 2

Show that if any five numbers from 1 to 8 are chosen, then two of them will add to 9.

Solution

- Pigeons :
- Pigeonholes :

- By PHP :

Odd part of a positive integer n

- Every positive integer n can be written as $n = 2^k m$, where m is odd and $k \geq 0$.
- For example, when $n \leq 20$

$1 = 2^0 \cdot 1$	$2 = 2^1 \cdot 1$	$3 = 2^0 \cdot 3$	$4 = 2^2 \cdot 1$
$5 = 2^0 \cdot 5$	$6 = 2^1 \cdot 3$	$7 = 2^0 \cdot 7$	$8 = 2^3 \cdot 1$
$9 = 2^0 \cdot 9$	$10 = 2^1 \cdot 5$	$11 = 2^0 \cdot 11$	$12 = 2^2 \cdot 3$
$13 = 2^0 \cdot 13$	$14 = 2^1 \cdot 7$	$15 = 2^0 \cdot 15$	$16 = 2^4 \cdot 1$
$17 = 2^0 \cdot 17$	$18 = 2^1 \cdot 9$	$19 = 2^0 \cdot 19$	$20 = 2^2 \cdot 5$

- This can be seen by simply factoring all powers of 2 (if any) out of n.
- In this case let us call m **the odd part of n**.

Odd part numbers of positive integer $n \leq 20$

m	n
1	$\{1,2,4,8,16\}$
3	$\{3,6,12\}$
5	$\{5,10,20\}$
7	$\{7,14\}$
9	$\{9,18\}$
11	$\{11\}$
13	$\{13\}$
15	$\{15\}$
17	$\{17\}$
19	$\{19\}$

- Let n_1 and n_2 be two numbers with the same odd part.
- We must have $n_1 = 2^{k_1}m$ and $n_2 = 2^{k_2}m$, for some k_1 and k_2. If $k_1 \geq k_2$, then n_1 is a multiple of n_2; otherwise, n_2 is a multiple of n_1.

Example 3

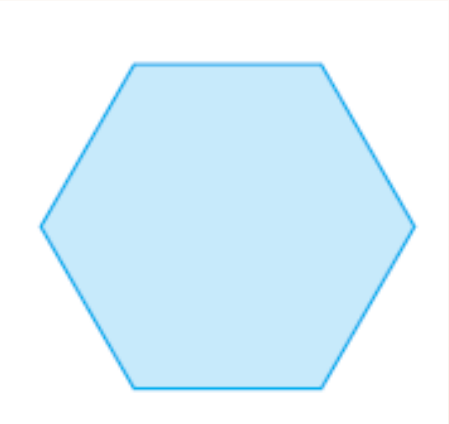
Show that if any 11 numbers are chosen from the set $\{1, 2, \ldots, 20\}$, then one of them will be a multiple of another.

Hint : The key to solving this problem is to create 10 or fewer pigeonholes in such a way that each number chosen can be assigned to only one pigeonhole, and when x and y are assigned to the same pigeonhole, we are guaranteed that either $x \mid y$ or $y \mid x$.

Solution

- Pigeons :
- Pigeonholes :
- By PHP :

Example 4



Consider a regular hexagon whose sides are of length 1 unit. Show that if any seven points are chosen in the area inside the shape, then two of them must be no farther apart than 1 unit.

Solution

- Pigeons :
- Pigeonholes :
- By PHP :

Example 5

Shirts numbered consecutively from 1 to 20 are worn by the 20 members of a bowling league.

When any three of these members are chosen to be a team, the league proposes to use the sum of their shirt numbers as **a code number for the team**.

Show that if any eight of the 20 are selected, then from these eight one may form at least two different teams having the same code number.

https://www.amazon.com/Numbers-consecutively-Transfer-Football-Baseball/dp/B07MYDFFLW

Example 5 (2)

Show that if any eight of the 20 are selected, then from these eight one may form at least two different teams having the same code number.

Solution

- Pigeons :
- Pigeonholes :
- By PHP :

The Extended Pigeonhole Principle

- **Theorem 2 :** **If n pigeons are assigned to m pigeonholes, then one of the pigeonholes must contain at least $\left\lfloor \frac{(n-1)}{m} \right\rfloor + 1$ pigeons.**

- **Proof** Use proof by contradiction.

  If each pigeonhole contains no more than $\left\lfloor \frac{(n-1)}{m} \right\rfloor$ pigeons, then there are at most $m \cdot \left\lfloor \frac{(n-1)}{m} \right\rfloor \leq m \cdot \frac{(n-1)}{m} = n - 1$ pigeons in all.

  This contradicts our hypothesis, so one of the pigeonholes must contain at least $\left\lfloor \frac{(n-1)}{m} \right\rfloor + 1$ pigeons.

Example 6

Show that if any 30 people are selected, then one may choose a subset of five so that all five were born on the same day of the week.

Solution

- Pigeons :
- Pigeonholes :
- By the extended pigeonhole principle (EPHP):

Example 7

Show that if 30 dictionaries in a library contain a total of $61{,}327$ pages, then one of the dictionaries must have at least $2{,}045$ pages.

Solution

- Pigeons :
- Pigeonholes :
- By EPHP :